<u>Assignment:- 1</u>

Number Systems conversions and 1’s and 2’s complement

There are primarily 4 number systems-

1.Decimal number system

2.Binary number system

3.Octal number system

4.Hexadecimal number system

1.Decimal number system-The Decimal number system is a widely used number system. This system is widely used in core mathematics, statistics, Physics and various other places that we may encounter in our day-to-day life. In this system the place of each value is determined by increasing powers of 10

Base-10

Digits Range-0-9

Conversion-To convert we multiply the digits of the number given with the base of the other number system and raise that base to powers starting from 0 until n-1 digit.

2.**Binary number system-**The Binary number system also known as the Machine Language is widely used in industries such as Computers, Electronics, Electrical etc. The Binary number systems consist of only 1s and 0s and the place value of each place is determined by the increasing powers of 2.

Base-2

Digits Range-0-1

Conversion-To convert a decimal number into binary we must keep on dividing the given number by 2 until we reach 0 Quotient and note all the remainders during each division. Then note down the combination of all remainders i.e 0s and 1s in reverse order to obtain its binary form.

Binary addition and subtraction rules-

Addition-

B1	B2	Sum	Carryover
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Subtraction-

B1	B2	Subtract	Carryover
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

3.**Octal number system-**The Octal number system is widely used in computing and programming, UNIX systems etc. The Octal number systems consist of 0-7 digits and the place value of each place is determined by the increasing powers of 8.

Base-8

Digits Range-0-7

Conversion-To convert a decimal number into octal we must keep on dividing the given number by 8 until we reach 0 Quotient and note all the remainders during each division. Then note down the combination of all remainders in reverse order to obtain its Octal form.

4.**Hexadecimal number system-**The Hexadecimal number system is widely used in computing and programming, representing memory addresses, Colour combinations etc. The Hexadecimal number systems consist of 0-9 digits and letters A-F (A=10, B=11, C=12, D=13, E=14, F=15) and the place value of each place is determined by the increasing powers of 16.

Base-16

Digits Range-0-9 and A-F (10-15)

Conversion-To convert a decimal number into Hexadecimal we must keep on dividing the given number by 16 until we reach 0 dividend and note all the remainders during each division. Then note down the combination of all remainders in reverse order to obtain its Hexadecimal form and properly write remainders >9 in the specified letter format.

1's Complement

The 1's complement of a number is basically just flipping of all the bits

Ex- 1's complement of 1101 is 0010

2's Complement

2’s complement of a binary number is the easiest and most common way to represent a negative number.It consists of 2 steps

1.Flip all bits

2.Add binary 1 to it

Ex- 2’s complement of 1101 is 0011

Subtraction Using 1’s Complement

To perform A – B:

Write A and B in binary (same number of bits).

Find the 1’s complement of B (the subtrahend).

Add this to A.

If there is a carry out, add it back to the result (end-around carry).

If there is no carry, the result is negative → take 1’s complement of the result to get its magnitude.

Example : 7 – 5 (4-bit)

A = 7 = 0111

B = 5 = 0101

Step 1: 1’s comp of B = 1010

Step 2: A + (1’s comp B) = 0111 + 1010 = (1)0001

Step 3: End-around carry → 0001 + 1 = 0010

Result = 2

Subtraction Using 2’s Complement

- To perform A – B:
- Write A and B in binary (same number of bits).
- Find the 2’s complement of B.
- Add this to A.
- If there is a carry out, discard it.
- The result is the correct answer (positive or negative).
- If the MSB is 0 → result is positive.
- If the MSB is 1 → result is negative (in 2’s complement form).

Example 1: 7 – 5 (4-bit)

A = 7 = 0111

B = 5 = 0101

Step 1: 2’s comp of B → 1011

Step 2: A + (2’s comp B) = 0111 + 1011 = (1)0010

Step 3: Discard carry → 0010

Result = 2

Questions-

Q1. What is the main advantage of using binary number system in computers?

- The Binary number system is used in computers and machines because it can easily be represented using switches such as on off replicating the 0s and 1s of Binary thus making it easy to understand and process.

Q2. What is the difference between a positional and non-positional number system?

- In a positional number system the value of a number depends on its position while in non-positional number system such as Roman numerals the value of a digit is fixed and doesn't depend on position.

Q. Convert Decimal numbers to Binary

1. 0.78125

Ans:-

- Fraction only (multiply by 2):
- $0.78125 \times 2 = 1.5625 \rightarrow$ bit = 1, new frac = 0.5625
- $0.5625 \times 2 = 1.125 \rightarrow$ bit = 1, new frac = 0.125
- $0.125 \times 2 = 0.25 \rightarrow$ bit = 0, new frac = 0.25
- $0.25 \times 2 = 0.5 \rightarrow$ bit = 0, new frac = 0.5
- $0.5 \times 2 = 1.0 \rightarrow$ bit = 1, new frac = 0 $\rightarrow$ stop
- Fraction bits (in order) = .11001
- Answer: $0.78125 = 0.11001_2$

2. 13.6875

Ans:-

- Integer part (13) — divide by 2 and record remainders:
- 13 ÷ 2 = 6, remainder 1
- 6 ÷ 2 = 3, remainder 0
- 3 ÷ 2 = 1, remainder 1
- 1 ÷ 2 = 0, remainder 1
- Read remainders bottom → top → integer bits = 1101 (which is 8 + 4 + 0 + 1 = 13).
- Fraction part (0.6875) — multiply by 2:
- 0.6875 × 2 = 1.375 → bit 1, frac 0.375
- 0.375 × 2 = 0.75 → bit 0, frac 0.75
- 0.75 × 2 = 1.5 → bit 1, frac 0.5
- 0.5 × 2 = 1.0 → bit 1, frac 0 → stop
- Fraction bits = .1011
- Answer: 13.6875 = 1101.1011_2

3. 192

- Ans:-
- Integer only. Division-by-2 remainders:
- 192 ÷ 2 = 96, r 0
- 96 ÷ 2 = 48, r 0
- 48 ÷ 2 = 24, r 0
- 24 ÷ 2 = 12, r 0
- 12 ÷ 2 = 6, r 0
- 6 ÷ 2 = 3, r 0
- 3 ÷ 2 = 1, r 1
- 1 ÷ 2 = 0, r 1
- Read bottom → top → 11000000

- (Equivalently: $192 = 128 + 64 \rightarrow$ bits for 2^7 and 2^6 set.)
- Answer: $192 = 11000000_2$

4. 653.625

Ans:-

- Integer part (653) — divide by 2:
- $653 \div 2 = 326$, r 1
- $326 \div 2 = 163$, r 0
- $163 \div 2 = 81$, r 1
- $81 \div 2 = 40$, r 1
- $40 \div 2 = 20$, r 0
- $20 \div 2 = 10$, r 0
- $10 \div 2 = 5$, r 0
- $5 \div 2 = 2$, r 1
- $2 \div 2 = 1$, r 0
- $1 \div 2 = 0$, r 1
- Read bottom $\rightarrow$ top $\rightarrow$ integer bits = 1010001101
- (Checks: $512 + 128 + 8 + 4 + 1 = 653$)
- Fraction part (0.625) — multiply by 2:
- $0.625 \times 2 = 1.25 \rightarrow$ bit 1, frac 0.25
- $0.25 \times 2 = 0.5 \rightarrow$ bit 0, frac 0.5
- $0.5 \times 2 = 1.0 \rightarrow$ bit 1, frac $0 \rightarrow$ stop
- Fraction bits = .101
- Answer: $653.625 = 1010001101.101_2$

Q. Convert Binary to Decimal

1. 110101

Ans:-

- Binary digits: 1 1 0 1 0 1
- (positions: 2^5, 2^4, 2^3, 2^2, 2^1, 2^0)
- $1 \times 2^5 = 32$
- $1 \times 2^4 = 16$
- $0 \times 2^3 = 0$
- $1 \times 2^2 = 4$
- $0 \times 2^1 = 0$
- $1 \times 2^0 = 1$
- Sum = 32 + 16 + 4 + 1 = 53

2. 101.1101

Ans:-

- Integer part (101_2): $= 1\times2^2 + 0\times2^1 + 1\times2^0 = 4 + 0 + 1 = 5$
- Fraction part ($.1101_2$):
- $1\times2^{-1} = 1/2 = 0.5$
- $1\times2^{-2} = 1/4 = 0.25$
- $0\times2^{-3} = 0$
- $1\times2^{-4} = 1/16 = 0.0625$
- Sum = 0.5 + 0.25 + 0 + 0.0625 = 0.8125
- Total = 5 + 0.8125 = 5.8125

3. 10101

Ans:-

- Digits: 1 0 1 0 1 (2^4 down to 2^0)
- $1 \times 2^4 = 16$
- $0 \times 2^3 = 0$
- $1 \times 2^2 = 4$
- $0 \times 2^1 = 0$
- $1 \times 2^0 = 1$
- Sum = 16 + 4 + 1 = 21

4. 1011.101

Ans:-

- Integer part (1011_2): $= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

  $= 8 + 0 + 2 + 1 = 11$
- Fraction part ($.101_2$):
- $1 \times 2^{-1} = 0.5$
- $0 \times 2^{-2} = 0$
- $1 \times 2^{-3} = 0.125$
- Sum = 0.5 + 0 + 0.125 = 0.625
- Total = 11 + 0.625 = 11.625

5.100101

Ans:-

- Digits: 1 0 0 1 0 1 (2^5 down to 2^0)
- $1 \times 2^5 = 32$
- $0 \times 2^4 = 0$
- $0 \times 2^3 = 0$

- $1 \times 2^2 = 4$
- $0 \times 2^1 = 0$
- $1 \times 2^0 = 1$
- Sum = $32 + 4 + 1 = 37$